悼念退潮之后：
基于多智能体仿真的逝者舆论形象塑造机制研究

49,930 –

10 30

一、研究背景

“ ” digital afterlife [1]

“ ”

West et al., 2021 TikTok Eriksson Krutrök, 2021 “ ”

2026 3 24

“ ”

49 930

1.

2.

3.

[1] https://www.elainekasket.com/insights/digital-afterlife-explained

□□□□□□

“ ” “ ”

Uwalaka 2023

Clancy 2024

2024

Adriaansen Smit 2025

Eriksson Krutrök 2021 “ ”

West 2021 2362

“ ” “ ”

“ ” “ ”

Shifman

Jenkins Nissenbaum

Bou-Franch Yus

“ — ”

“ ” “ ” Metzler Garcia 2024

2024

Huszár 2022

Hosseinmardi 2024

–

LLM

□□□□□□

Spiral of Silence Theory Curated Flows

Noelle-Neumann 1974

Sohn

Geidner 2016

Thorson Wells 2016

B D R R

D

B R

B R

Python “ ” 2026 1 5 49,930 16,602

33,328

1 2

3,315 13 46,602

1

	“ ”“ ” # # #
	/ / / / / / / / / /App/ / /
	/ /
	/ / / / / / / / /
	UI AI OCR

400

84.0% Cohen's κ 0.799 Bootstrap 95% CI [0.753, 0.842] F1 0.96 0.89 0.86 F1=0.66

2

Po	84.0% 336/400	AI
Pe	20.5%	
Cohen's κ	0.799	
κ 95%	[0.753, 0.842]	bootstrap B=5000 seed=20261010

3

		(PPV)		**F1**
	136	98.4%	93.4%	0.96
	58	65.0%	67.2%	0.66
	78	81.2%	66.7%	0.73
	54	75.8%	92.6%	0.83
	48	87.8%	89.6%	0.89
	26	78.1%	96.2%	0.86

RQ1

~~49,930~~ 49,930 46,602

1a ,

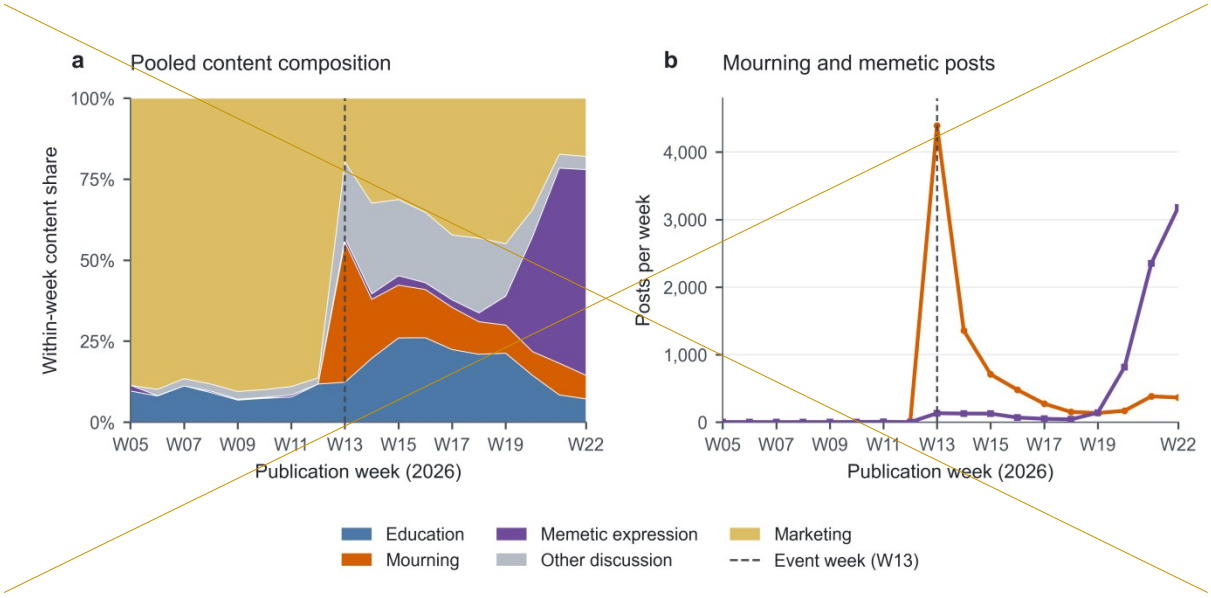


Fig. 1a

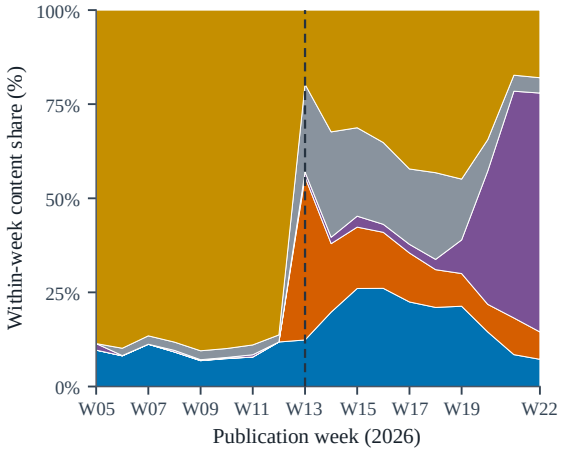
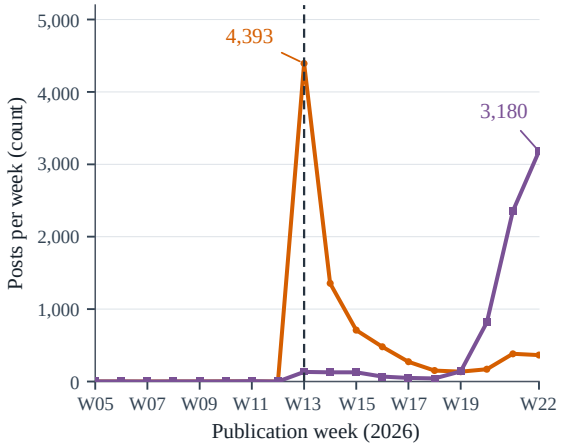


Fig. 1b



Education Mourning Memetic expression Other discussion Marketing

---- W13: Zhang Xuefeng's death (24 Mar 2026)

1

1a 1b 2026 ISO W05–W22 W13 2026 3 24 1b 49,930 46,602 3,315 13

1a

1b W13

W13_2026_3_24 4,000

W19 W20–W22 W22 3,200 63.5%

2

W13

W22 82.4% 19.3%

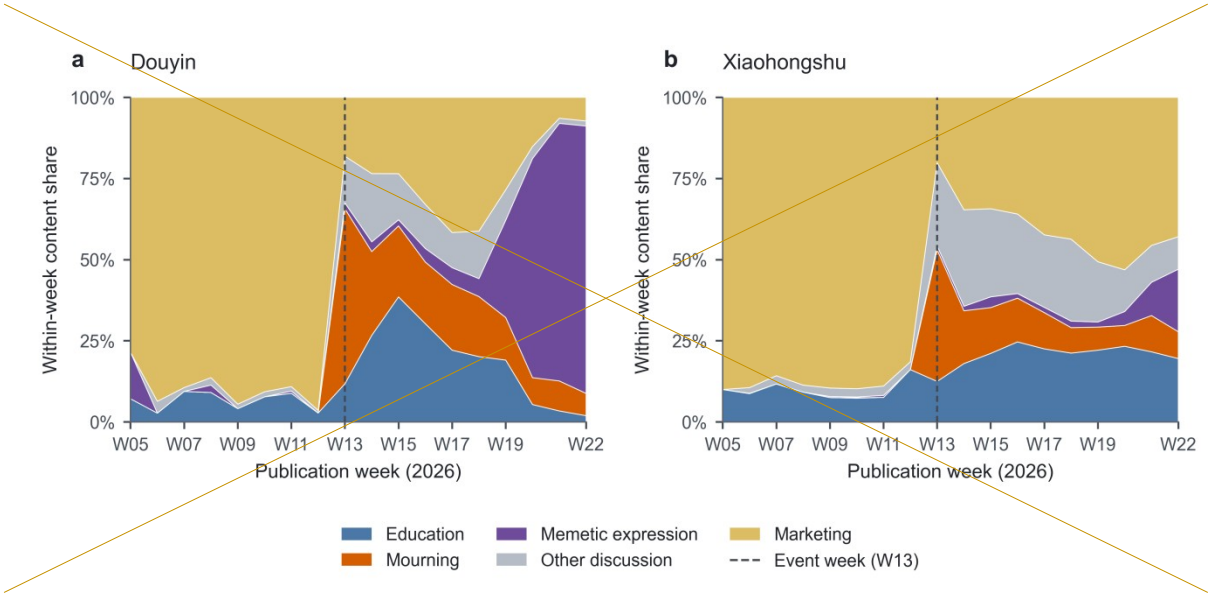
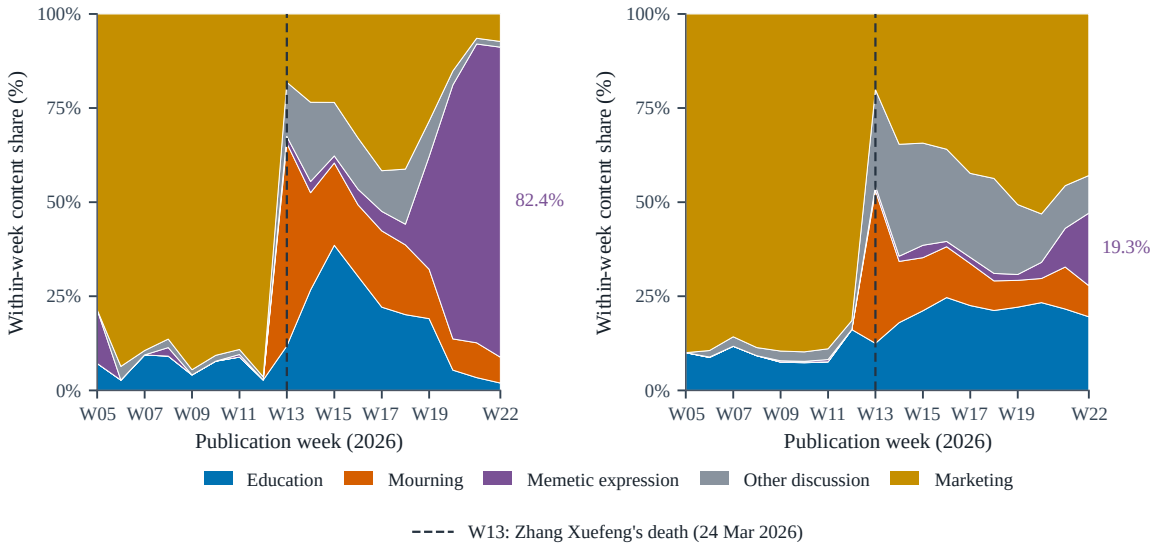


Fig. 2a Douyin **Fig. 2b** Xiaohongshu



2

2a 2b 0–100%

W13 2026 3 24 2026 W05–W22

1 46,602 15,186 31,416

W13

W19 W20–W22 W22 82.4%

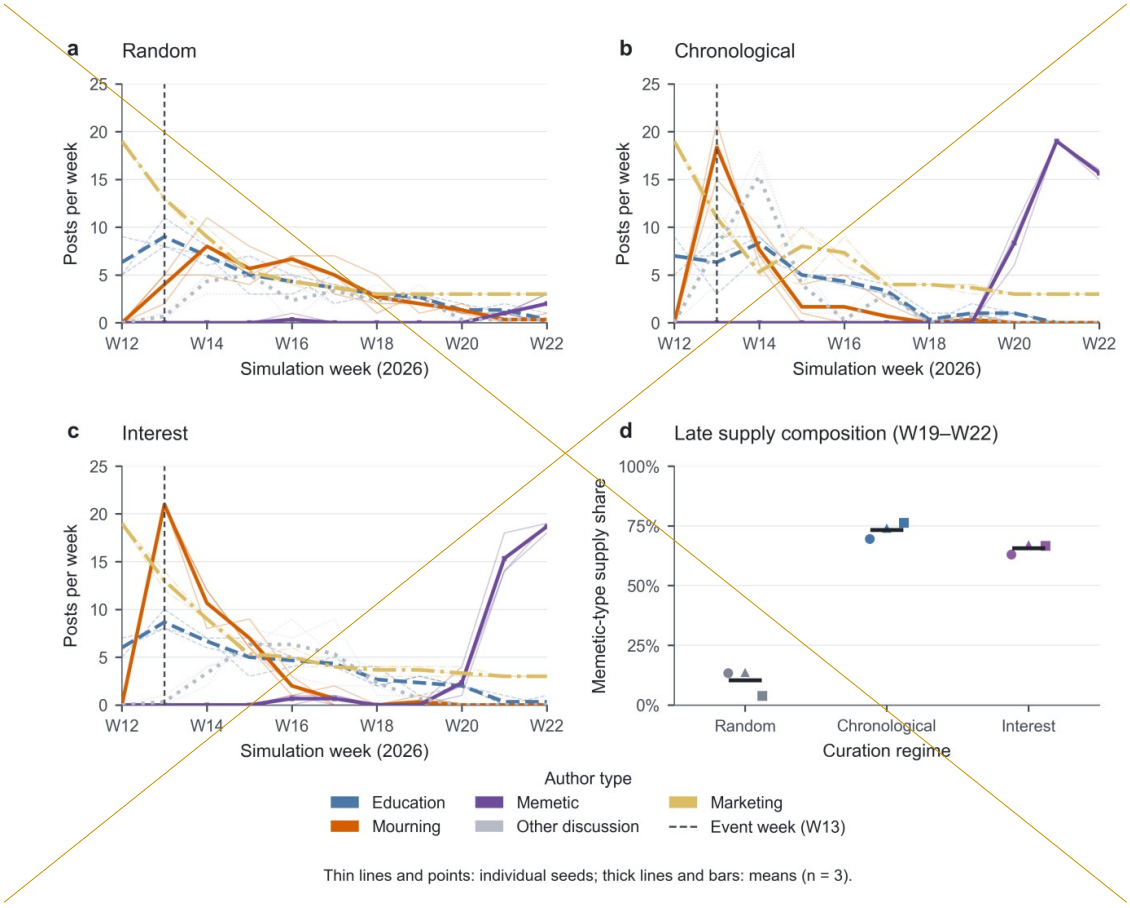
W13 W22 19.3%

63.1

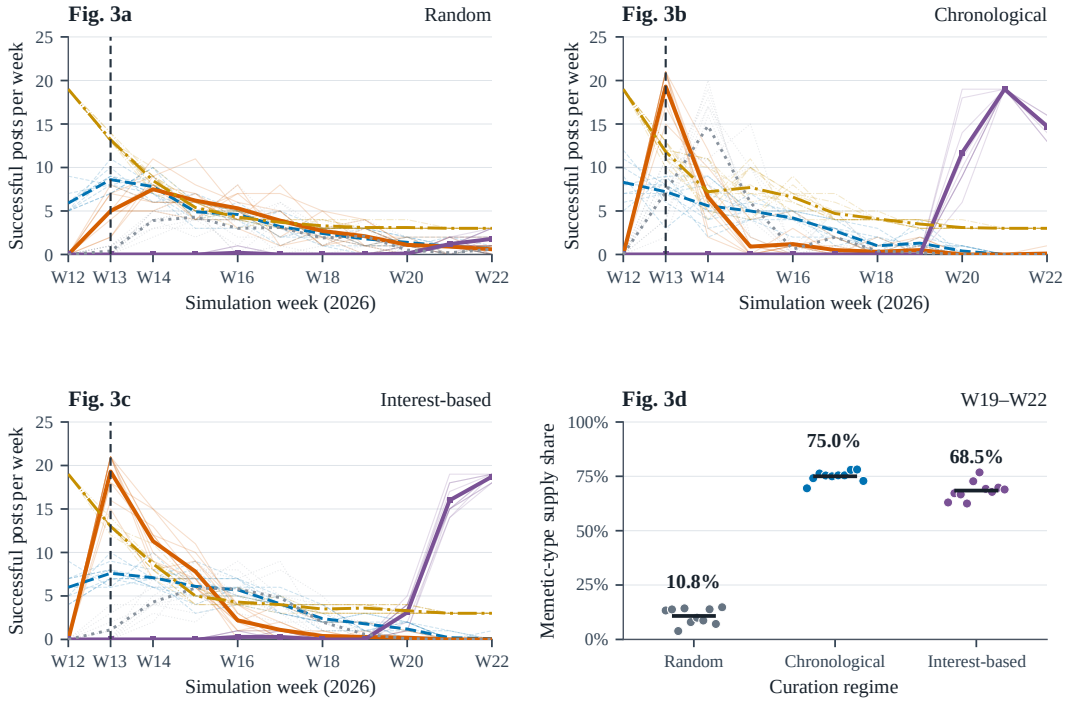
——

□□□

W13
W14



Thin lines and points: individual seeds; thick lines and bars: means (n = 3).



Thin lines / points: individual runs; thick lines / black bars: means.
n = 10 seeds per regime; no inferential error bars.

3

3a–c W12–W22 10

W13 2026 3 24 3d W19–W22 10

10.8% 75.0% 68.5% seed 0–9 30

3 7

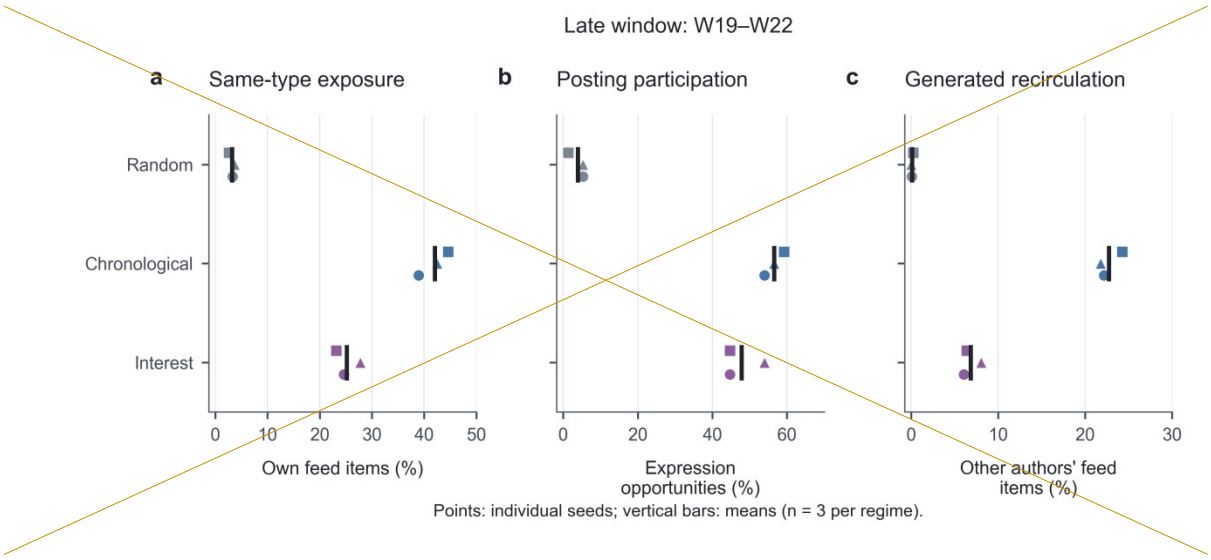
W19–W22

3.0 43.0 36.3 3.1 45.4 37.8 10.3% 73.3% 65.6% 10.8% 75.0% 68.5%

W21 W22 W21 W22

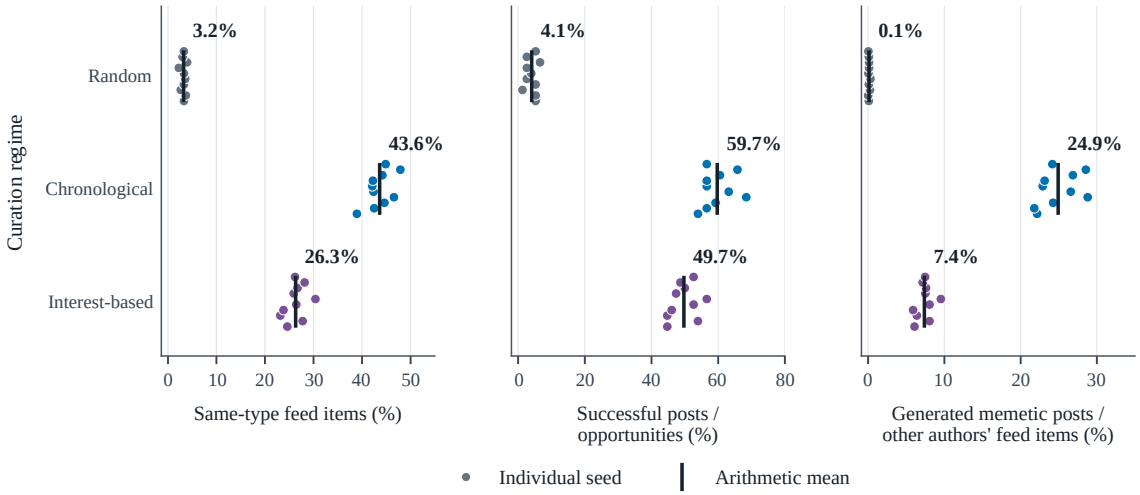
Thorson Wells 2016

Trilling 2024



W19–W22 · n = 10 seeds per regime

Fig. 4a **Fig. 4b** **Fig. 4c**



4

4a W19–W22 4b 4c

10 4a 760 4b

19 4 76 4c 3,240 4a 4c

3.2% 43.6% 26.3% 4.1% 59.7% 49.7%

0.1% 24.9% 7.4%

W19–W22

42.0% 25.2% 43.6% 26.3% 3.2% 4a

56.6%

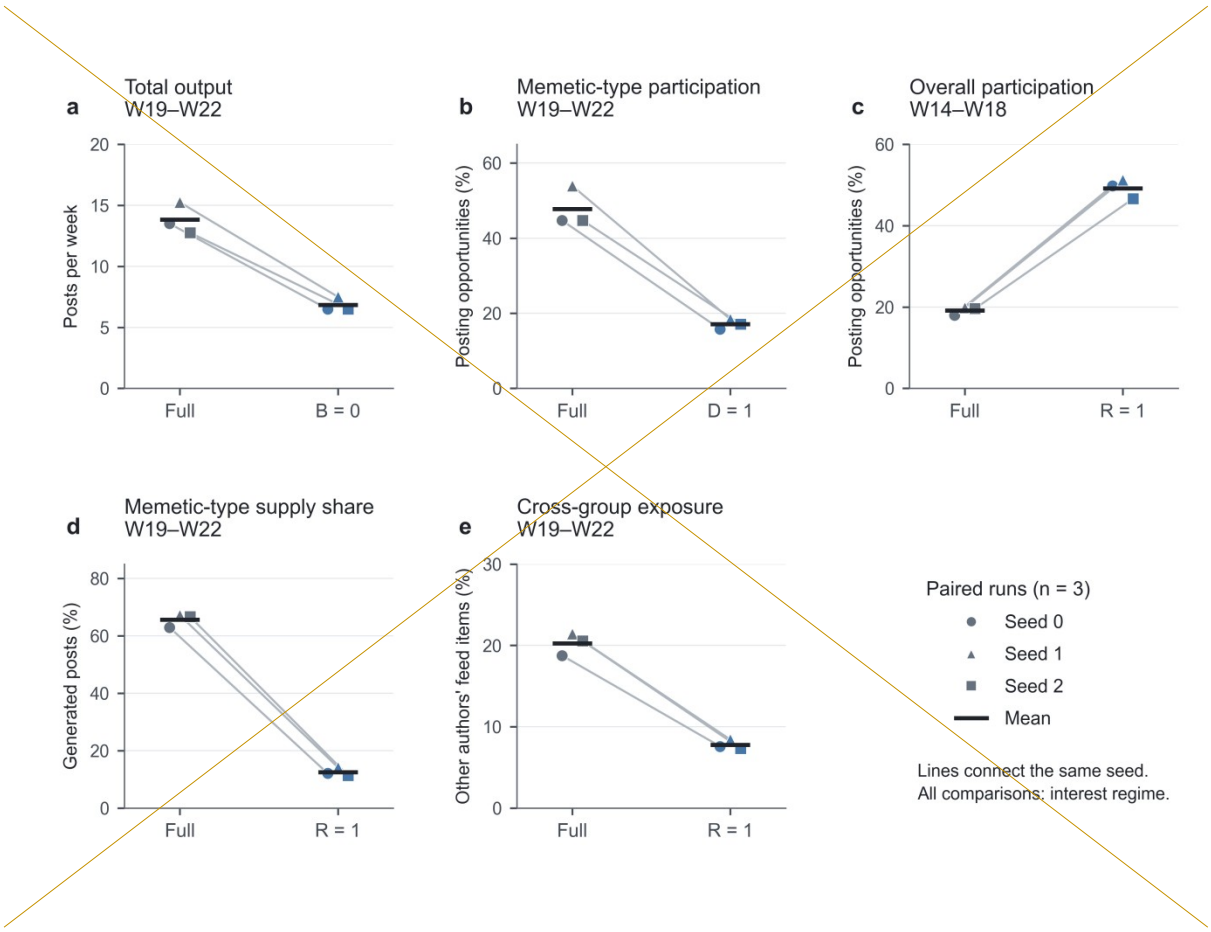
47.8% 59.7% 49.7% 3.9% 4.1% 4b

W19—W22

~~22.7% 6.8%~~24.9% 7.4% 0.1% 4c

~~B D~~

~~R 5~~



5

B=0 W19—W22

13.8 6.8

D=1

W19—W22 47.8% 17.1%

R=1 W14—W18

19.1% 49.2% W19—W22 65.6% 12.6%

20.2%

7.8%

六、讨论与结论

B D R

□□□□

Adriaansen, R. J., & Smit, R. (2025). Collective memory and social media. *Current Opinion in Psychology, 65*, Article 102077. https://doi.org/10.1016/j.copsyc.2025.102077

Bou-Franch, P., & Yus, F. (2025). Humour in times of mourning? A contrastive analysis of memes. *European Journal of Humour Research, 13*(2), 1–25. https://doi.org/10.7592/EJHR.2025.13.2.1070

Cabrera, B., Ross, B., Röchert, D., Brünker, F., & Stieglitz, S. (2021). The influence of community structure on opinion expression: An agent-based model. *Journal of Business Economics, 91*(9), 1331–1355. https://doi.org/10.1007/s11573-021-01064-7

Candia, C., Jara-Figueroa, C., Rodriguez-Sickert, C., Barabási, A.-L., & Hidalgo, C. A. (2019). The universal decay of collective memory and attention. *Nature Human Behaviour, 3*(1), 82–91. https://doi.org/10.1038/s41562-018-0474-5

Clancy, L. (2024). #MournHub and @GrieveWatch: Mediating monarchy and mourning in the digital age. *International Journal of Cultural Studies, 27*(3), 359–376.

Eriksson Krutrök, M. (2021). Algorithmic closeness in mourning: Vernaculars of the hashtag #grief on TikTok. *Social Media + Society, 7*(3), 1–12.

. (2024). . , 45(2), 150–160.

Hosseinmardi, H., Ghasemian, A., Rivera-Lanas, M., Horta Ribeiro, M., West, R., & Watts, D. J. (2024). Causally estimating the effect of YouTube's recommender system using counterfactual bots. *Proceedings of the National Academy of Sciences, 121*(8), Article e2313377121. https://doi.org/10.1073/pnas.2313377121

Huszár, F., Ktena, S. I., O'Brien, C., Belli, L., Schlaikjer, A., & Hardt, M. (2022). Algorithmic amplification of politics on Twitter. *Proceedings of the National Academy of Sciences, 119*(1), Article e2025334119. https://doi.org/10.1073/pnas.2025334119

Jenkins, H. (2006). *Convergence culture: Where old and new media collide*. New York University Press.

, , & . (2024). —— . , 46(4), 90–112.

Matthes, J., Knoll, J., & von Sikorski, C. (2018). The "spiral of silence" revisited: A meta-analysis on the relationship between perceptions of opinion support and political opinion expression. *Communication Research, 45*(1), 3–33. https://doi.org/10.1177/0093650217745429

Metzler, H., & Garcia, D. (2024). Social drivers and algorithmic mechanisms on digital media. *Perspectives on Psychological Science, 19*(5), 735–748. https://doi.org/10.1177/17456916231185057

Nissenbaum, A., & Shifman, L. (2017). Internet memes as contested cultural capital: The case of 4chan's /b/ board. *New Media & Society, 19*(4), 483–501.

Noelle-Neumann, E. (1974). The spiral of silence: A theory of public opinion. *Journal of Communication, 24*(2), 43–51. https://doi.org/10.1111/j.1460-2466.1974.tb00367.x

Shifman, L. (2013). *Memes in digital culture*. MIT Press.

Sohn, D., & Geidner, N. (2016). Collective dynamics of the spiral of silence: The role of ego-network size. *International Journal of Public Opinion Research, 28*(1), 25–45. https://doi.org/10.1093/ijpor/edv005

Thorson, K., & Wells, C. (2016). Curated flows: A framework for mapping media exposure in the digital age. *Communication Theory, 26*(3), 309–328. https://doi.org/10.1111/comt.12087

Trilling, D. (2024). Communicative feedback loops in the digital society. *Weizenbaum Journal of the Digital Society, 4*(2). https://doi.org/10.34669/wi.wjds/4.2.4

Uwalaka, T. (2023). Connective mourning: The case of mourning and memorialization practices on X. *OMEGA—Journal of Death and Dying, 92*(3), 1505–1528. https://doi.org/10.1177/00302228231208112

West, R., Leskovec, J., & Potts, C. (2021). Postmortem memory of public figures in news and social media. *Proceedings of the National Academy of Sciences, 118*(38), Article e2106152118. https://doi.org/10.1073/pnas.2106152118

Wu, F., & Huberman, B. A. (2007). Novelty and collective attention. *Proceedings of the National Academy of Sciences, 104*(45), 17599–17601. https://doi.org/10.1073/pnas.0704916104